537. Adolf Baeyer and Victor Villiger: Action of Caro's Reagent on Ketones.

[Preliminary communication from the Chemical Laboratory of the Academy of Sciences at Munich.]

(Received December 27.)

English translation of Ber. dtsch. chem. Ges. 32 (1899) 3625–3633. Structural formulas are the originals; their labels remain in the original notation. Article No. 538 is a separate paper, not included.

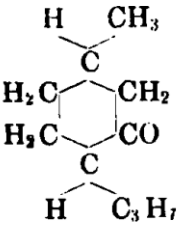
H. Caro[1]) found last year that the salts of persulfuric acid, when brought together with concentrated sulfuric acid, undergo a change still unknown in its nature, such that the solution — afterwards diluted with water — cleanly oxidizes aniline to nitrosobenzene. Bamberger and Tschirner[2]) subsequently succeeded in moderating the oxidation of aniline still further and halting it at the formation of phenylhydroxylamine.

While engaged in work on the cleavage of the ring-shaped ketones of the terpene group, we were drawn to these observations and found that Caro's reagent is an agent acting just as specifically on ketones, one which allows a series of previously unachievable tasks to be solved with playful ease.

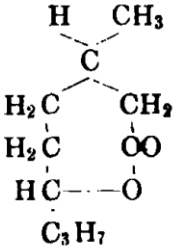
In what follows we publish the first results of this investigation, with the kind permission of Mr. Caro, and we hope to be able to fill the remaining gaps in the shortest time.

Caro's reagent is an oxidizing agent which, in the manner of its action, resembles hydroxylamine. Thus ethyl alcohol is not appreciably attacked by it, while most ketones are oxidized. This similarity, however, goes still further, in that the oxidation products of many ketones possess a constitution corresponding to the rearrangement of oximes discovered by Beckmann.

Thus menthone and tetrahydrocarvone are cleanly converted by the reagent into the corresponding ε-lactones, of which the first was described in the preceding paper.

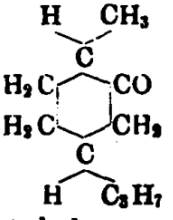


Menthon

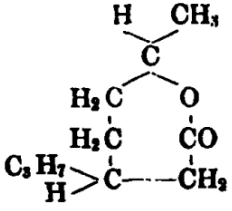


ε-Lacton.

Tetrahydrocarvone behaves likewise, giving the corresponding ε-lactone:

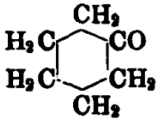


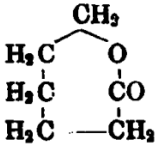
Tetrahydrocarvon



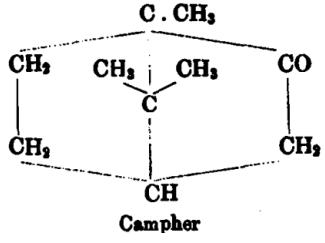
ε-Lacton.

Ketocyclohexane too behaves in the same way, according to a preliminary orientation:

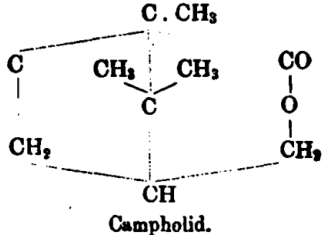




Camphor behaves similarly as well, yielding in abundant amount the campholide — the phthalide of camphoric acid — which Haller[1]) obtained by reduction of camphoric anhydride.



Campher

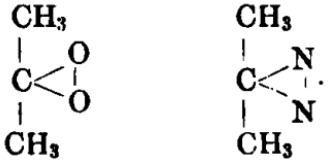


Campholid.

Alongside the campholide, this reaction also gives a more oxygen-rich, beautifully crystallized lactone of composition

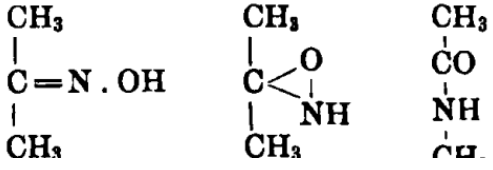
$C_{10}H_{16}O_4$, which appears to be still unknown and has not yet been examined more closely.

To obtain insight into the nature of the process that leads from the ketone to the lactone, we treated acetone with the reagent. There forms at once a beautifully crystallized, very volatile oxidation product which, on analysis, showed the same composition as the acetone peroxide of Wolffenstein[2]) and also possesses similar properties, but differs from it by a melting point more than 30° higher. The molecular-weight determination will show whether we are dealing here with a polymeric or a stereoisomeric body; the explosiveness and the chemical indifference of the substance leave no doubt that it, like Wolffenstein's compound, is a polymer of the simple acetone peroxide. Although this simple acetone peroxide — which probably stands to the polymer as acetaldehyde to paraldehyde or metaldehyde — could not hitherto be isolated, one will hardly go wrong in ascribing to it a constitution corresponding to the aliphatic diazo compounds:

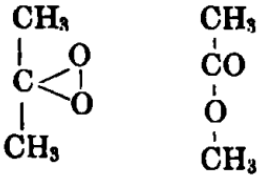


Likewise it is very probable that, just as the formation of the polymeric acetone peroxide should be preceded by that of a simple one, so too the cyclic ketones are converted in the first instance into peroxides, which for some unknown reason do not rearrange into a polymeric modification but into lactones. This admittedly only hypothetical rearrangement corresponds exactly to the Beckmann rearrangement of the oximes, if one assumes that it is preceded by conversion into the isomeric form corresponding to the peroxide:

Beckmann rearrangement.



Rearrangement of the hypothetical peroxide.



It is to be hoped that these questions can be answered by the already begun investigation of the behavior of more carbon-rich open-chain ketones. Likewise we have drawn the accessible cyclic ketones, such as suberone etc., into the scope of our investigation, and will report shortly on the results obtained.

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[1]) Zeitschr. f. angewandte Chemie 1898, 845. [2]) Diese Berichte 32, 1676.
(camphor) [1]) Diese Berichte 29, R. 221. [2]) Diese Berichte 28, 2265.

EXPERIMENTAL

The Reagent

Caro's reagent was prepared by triturating potassium persulfate with concentrated sulfuric acid, or with sulfuric acid diluted by one molecular equivalent of water. It was used in three forms:

1. Dry reagent. 11 g of concentrated sulfuric acid were triturated in a mortar with 10 g of potassium persulfate; after standing for 10 minutes, 30 g of powdered potassium sulfate were added and the mixture triturated again until a completely dry powder had formed. In this form the reagent is stable when kept from moisture and is the most useful for most purposes.

2. Liquid reagent is obtained by triturating potassium persulfate with three times its weight of concentrated sulfuric acid treated with one molecular equivalent of water. This is best suited for the oxidation of camphor.

3. Dilute reagent. 11 g of concentrated sulfuric acid and 10 g of potassium persulfate are triturated and brought to 50 cc with ice.

Menthone

Dry reagent, prepared from 38 g of persulfate, was cooled with ice-water in an Erlenmeyer flask, and 15 g of menthone added

gradually with stirring, taking care that the temperature never rose above 20°. After standing 24 hours the brownish mass was treated with water, extracted several times with ether, and the ethereal solution shaken out first with bicarbonate, then with dilute caustic soda. The alkali-soluble brown substances were not examined more closely; the 9 g remaining in the ether were subjected to fractional distillation. At 15 mm pressure the following passed over:

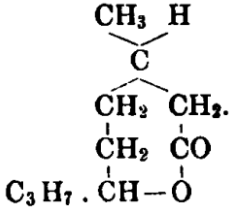
I.	95–120°	2.8 g
II.	120–137°	1.6 g
III.	137–140°	4.3 g

No residue remained. The first fraction consisted of unattacked menthone; the second solidified partly in the cold, the third completely. The crystals were freed of adhering oil on porous clay and then, after recrystallization from wood spirit, showed the melting point of the ε-lactone described by Seuffert in the preceding paper, 46–48° (Seuffert 46–47°). The analysis likewise gave agreeing figures:

0.1869 g subst.: 0.4829 g CO_2, 0.1781 g H_2O.

$C_{10}H_{18}O_2$ Calc. C 70.59, H 10.59. Found C 70.47, H 10.59.

The body is therefore identical with the high-melting ε-lactone obtained on distillation of 2,6-dimethyloctan-3-olic acid, and possesses the formula:



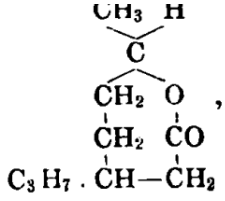
This was further confirmed by conversion of the lactone into the hydroxy acid and the keto acid. The hydroxy acid crystallizes from hot water in long leaves, m.p. 65–66° (Seuffert 65°). Of its salts, the silver, copper, zinc and lead salts crystallized. The keto acid obtained from it is liquid and gave an oxime melting, after recrystallization, at 100–102°; Baeyer and Oehler found 103°.

Tetrahydrocarvone

The dry reagent prepared from 50 g of persulfate was treated with 20 g of tetrahydrocarvone in exactly the manner just described, and the mass worked up in precisely the same way. 15 g of an alkali-insoluble oil were obtained, which was subjected to fractional distillation at 21 mm pressure.

I.	113–135°	2.9 g
II.	135–155.5°	3.2 g
III.	155.5–156°	6.8 g
IV.	156–158°	0.9 g

No residue. The first fraction was strongly contaminated with tetrahydrocarvone. The third consisted of almost pure lactone and dissolved completely on boiling with caustic soda, but did not solidify in a freezing mixture. The analysis of the third fraction led to the formula of the ε-lactone of 5-isopropylheptan-2-olic acid, which is still unknown:



0.1828 g subst.: 0.4689 g CO_2, 0.1753 g H_2O.

$C_{10}H_{18}O_2$ Calc. C 70.59, H 10.59. Found C 69.96, H 10.66.

By saponification the lactone passes into a recrystallizable hydroxy acid whose silver salt forms readily soluble needles in water. By treatment with the Beckmann mixture it passes into the keto acid, which Oehler[1]) obtained by direct oxidation of tetrahydrocarvone. The latter boils at 180–185° under 12 mm and, on seeding with Oehler's acid, solidified completely to a crystalline mass melting at 38° (Oehler 40°). The phenylhydrazone melts at 102–104° (Oehler 102°), the recrystallized oxime at 74° (Oehler 75–78°), the semicarbazone, after repeated recrystallization from alcohol, at 145–152° (Oehler 152–153°). From all of this the identity of the present acid with Oehler's keto acid follows, and thence the constitution assumed above for the hydroxy acid and its lactone.

Finally, be it noted that the lactone, alongside acids, also forms when the tetrahydrocarvone is shaken for 24 hours with the dilute reagent.

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[1]) Diese Berichte 29, 27.

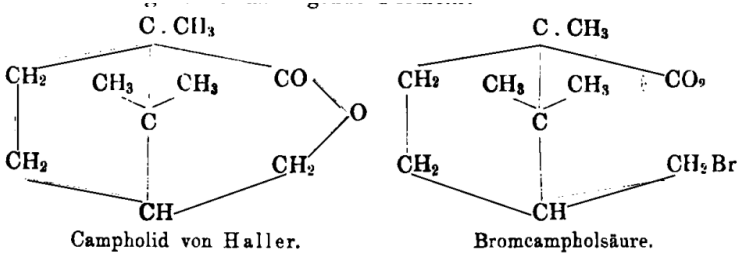
Camphor

The oxidation of camphor succeeds best with a mixture of absolute alcohol and the liquid reagent, with gradual addition of camphor dissolved in benzine. The addition of alcohol is, however, not necessary, as the following experiment shows:

To 1200 g of concentrated sulfuric acid treated with 1 molecular equivalent of water, 400 g of potassium persulfate were added, and into the cooled mixture a solution of 50 g of camphor in 75 g of benzine was dripped, with care that the temperature never rose above 20°. At each addition the mass turns brown and becomes colorless again after some shaking. The whole operation took, in 2 portions, 1½ hours. After this time the camphor had entirely disappeared, and so had the reacting substance, since the faintly brownish liquid scarcely separated any more iodine from potassium iodide solution. The whole mixture was then brought to 5 L with ice, extracted several times with ether, and the ether washed with sodium bicarbonate and dilute caustic soda. After distilling off the ether, 12.3 g of a camphor-like mass were obtained, which on boiling with distilled hydriodic acid changed into distinct crystals, whereas camphor under these conditions gives an oil. After it had been established that this iodine-containing compound regenerates the original substance on treatment with alkalis, the crude product was converted, for purification, into the likewise crystallizing bromine compound.

When the crude product was poured over with 150 g of glacial-acetic-acid/hydrogen bromide, the initially clear solution began after some hours to deposit leafy crystals, whose amount no longer increased after 2 days. After washing with ice-water and drying in vacuum, their weight was 10.4 g. The substance was finally purified by washing with a little ether. It is an acid and dissolves clearly in sodium carbonate, but the solution soon becomes turbid and deposits an indifferent body, identical with the campholide obtained by Haller through reduction of camphoric anhydride.

From this the following formulas result for the campholide and the bromine-containing substance:



Bromocampholic acid

Bromocampholic acid crystallizes from ether or ethyl acetate in large plates. It melts at 177° with gas evolution.

0.2012 g subst.: 0.3533 g CO_2, 0.1246 g H_2O.

$C_{10}H_{17}BrO_2$ Calc. C 48.19, H 6.83. Found C 47.89, H 6.88.

The somewhat too low carbon content found is explained by the substance still containing a trace of bromine.

Campholide of Haller

To prepare this substance, the ethereal solution of bromocampholic acid was shaken with alcoholic potash until a sample was bromine-free, and the ether then distilled off after washing with water. The remaining crystalline mass was recrystallized from warm ligroin, from which it separates in long, lustrous needles resembling sublimed phthalic anhydride. In boiling water it is sparingly soluble and separates on cooling in six-sided platelets. The taste is somewhat burning, as Haller also states. The melting point was found at 210–211° (Haller 211°). The substance is volatile with water vapor and possesses a very faint camphor odor. On oxidation with alkaline permanganate at water-bath temperature it cleanly yields camphoric acid.

0.1844 g subst.: 0.4800 g CO_2, 0.1573 g H_2O.

$C_{10}H_{16}O_2$ Calc. C 71.43, H 9.52. Found C 70.99, H 9.48.

New oxidation product of camphor

Since in the oxidation of camphor much more oxygen is consumed than is required for formation of the campholide, a

higher oxidation product was sought in the mother liquors. There was indeed found in them a product, hard to extract with ether and readily soluble in water, with lactone properties. It crystallizes from water in large plates, m.p. 189–191°, and has the composition $C_{10}H_{16}O_4$.

0.1700 g subst.: 0.3727 g CO_2, 0.1209 g H_2O.

$C_{10}H_{16}O_4$ Calc. C 60.00, H 8.00. Found C 59.79, H 7.90.

A c e t o n e

The dry reagent from 25 g of persulfate was poured over with 50 g of ether, cooled to 10°, and 5 g of acetone added all at once. No coloration occurs; the temperature rises gradually and is held at about 15° by cooling. After addition of ice the ether is lifted off, washed with bicarbonate and distilled off. There remains a dry white crystalline mass, amounting to 4.2 g, which was driven over with water vapor for purification. Dried in vacuum, the substance shows the composition of an acetone peroxide, $C_3H_6O_2$.

0.1911 g subst.: 0.3440 g CO_2, 0.1427 g H_2O.

$C_3H_6O_2$ Calc. C 48.65, H 8.11. Found C 49.09, H 8.30.

The body is extraordinarily volatile with water vapor and even with ether vapor, very slightly soluble in water, and not readily soluble in the other solvents either. It crystallizes from warm ethyl acetate in right-angled prisms, m.p. 132–133°. It explodes very violently on impact and on rubbing, but can be sublimed in small quantities in a test tube. Toward reagents it is very stable. The substance thus possesses, except for the melting point, the greatest similarity to the acetone peroxide which Wolffenstein prepared by the action of concentrated hydrogen peroxide on acetone. To determine whether the presence of the acidic potassium sulfate influences the formation of this acetone peroxide, a 10 percent solution of hydrogen peroxide was saturated with acid potassium sulfate and some acetone added. After standing 24 hours, a considerable amount of Wolffenstein's acetone peroxide had formed, m.p. 94–95°. The formation of the higher-melting peroxide therefore rests on a specific action of Caro's reagent.

Oxidation of Terpineol

5 g of terpineol were shaken, with a few drops of ether, with 50 cc of the dilute reagent. When the terpineol had disappeared after quite a short time, the liquid was neutralized with potassium carbonate, brought to dryness, and extracted with alcohol. In good yield a syrupy product boiling between 185–190° at 15 mm was obtained, which proved identical with the trioxyhexahydrocymol obtained by Wallach[1]) through oxidation with permanganate, since on oxidation with chromic acid it passed quantitatively into the methyl ketone of homoterpenylic acid[2]), $C_{10}H_{16}O_3$, discovered by Wallach. The m.p. was at 63–64° (Wallach 62–63°).

The reagent therefore does not act specifically on the double bond in terpineol, but just like dilute cold permanganate solution.

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[1]) Ann. d. Chem. 275, 150. [2]) Ann. d. Chem. 275, 153.